1 Physics Period 1, Week 1

1. The power output of a typical large power plant is 1000 megawatts (MW). Express this result in

   (a) W, (b) kW, (c) GW.

2. The diameter of a hydrogen atom is about 0.1 nm, and the diameter of a proton is about 1 fm. How many times bigger than a proton is a hydrogen atom?

3. Lake Baikal in Siberia holds the world's largest quantity of fresh water, about 14 Eg. How many kilograms is that?

4. A hydrogen atom is about 0.1 nm in diameter. How many hydrogen atoms lined up side by side would make a line 1 cm long?

5. A car is moving at 35 mi/h. Express its speed in (a) m/s and (b) ft/s.

6. How many cubic centimeters are in a cubic meter?

7. A gallon of paint covers 350 ft^2. What's its coverage in $\frac{m^2}{L}$?

8. One $\frac{m}{s}$ is how many $\frac{km}{h}$?

9. To see why it's important to carry more digits in intermediate calculations, determine $(\sqrt{3})^3$ to three significant figures in two ways:

   (a) Find $\sqrt{3}$ and round to three significant figures, then cube and again round,

   (b) Find $\sqrt{3}$ to four significant figures, then cube and again round to three significant figures.

10. You're driving into Canada and trying to decide whether to fill your gas tank before or after crossing the border. Gas in the United States costs \$2.97/gallon, in Canada C\$0.94/L, and Canadian dollar is worth \$0.7898. Where should you fill up?

11. In the 1908 London Olympics, the intended 26-mile marathon was extended 385 yards to put the end in front of the royal reviewing stand. This distance subsequently became standard. What is the marathon distance in kilometers, to the nearest meter?